Project Runway: The Power Ring

Our Idea

For Project Runway, we decided to create a wearable ring inspired by the Blue Lantern rings from DC Comics. We wanted to make something that looked like it came from a comic book but could also respond to the person wearing it. Our idea was to have the ring light up when someone closed their fist. We also wanted to give it a limited amount of charge, so the light would gradually dim as it was being used. Once the charge ran out, the ring would turn off and automatically recharge over a short period of time. We thought this would make the ring more interesting than just having an LED turn on and off. We wanted it to feel like the wearer was actually using the ring's power.

How the Ring Works

The ring uses a blue LED connected to an Arduino through wires running down the wearer's arm. To activate it, the wearer closes their fist, bringing two conductive foil contacts together. When the contacts touch, the Arduino detects the connection and activates the LED. The program keeps track of the ring's charge, which decreases while the ring is being used. As the charge goes down, the LED gradually becomes dimmer. Once the charge runs out, the LED turns off and the ring recharges automatically over a short period of time. We also used a glove to keep the foil contacts insulated. Without it, the contacts could accidentally activate when we didn't want them to.

Materials

- Arduino Uno
- Breadboard
- Blue LED
- Transistor
- 10K resistor and other resistors
- Jumper wires and longer insulated wires
- Conductive foil
- Rubber bands
- Blue paint
- 3D-printed ring
- USB cable
- Tinkercad

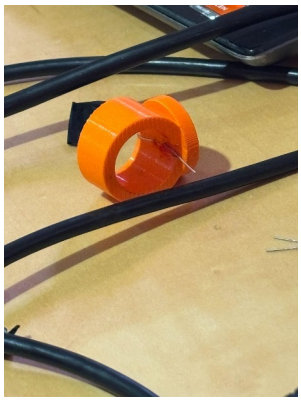
Planning the Project

We started by figuring out how we wanted the ring to work. The main idea was that closing a fist would activate the light, and keeping it active would gradually drain its charge. Before building everything, we worked on pseudocode to plan out how the activation, power drain, and recharge

functions would work. This helped us figure out what the Arduino needed to do before we started putting the final program together. We needed the Arduino to recognize when the contacts were connected, decrease the charge while the ring was being used, and allow it to recharge after running out.

Designing the Ring

We started by looking at different versions of the Blue Lantern ring from the comics. Since there are several designs, we had to decide which one would work best for our project. Adam sketched out the initial design on paper and figured out how the LED and wires could fit inside it. Instead of downloading a premade 3D model, Adam taught himself how to use Autodesk Fusion and created the ring himself. He also completed the required 3D-printing training at the General Makerspace. We originally wanted to print the ring in blue, but the queue was too long. We ended up printing it in orange instead and planned to paint it blue afterward.

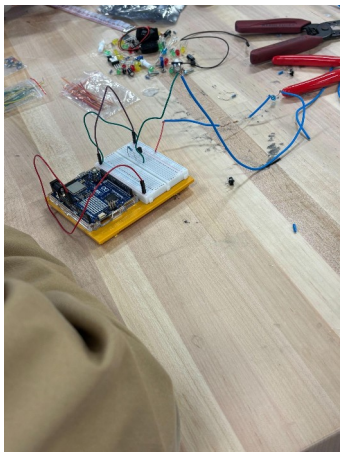


Our ring after being 3D printed.

The print also needed some adjustments. The channel meant for the wires was too narrow, so Adam had to manually carve out more space while removing the supports.

Building the Circuit and Programming

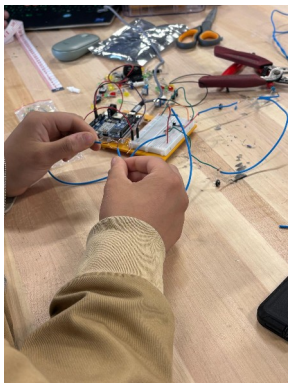
While the ring was being designed, we also started working on the electronics. Sam began with the idea of using a transistor to control the LED's fading effect. We used a Tinkercad example as a starting point and made changes to fit our project. For the early version, we used a pushbutton to represent the foil contacts. This allowed us to test the circuit before connecting everything to the physical ring. Sam wrote an initial version of the code, but parts of it were changed as we continued developing the circuit. We switched to using analogRead() at A0 to detect the input and analogWrite() to control the LED's brightness. One problem we ran into was that the charging system wasn't resetting correctly. After troubleshooting, we added a 10K pull-down resistor between the A0 input and ground. This helped stabilize the input and fixed the issues we were having when using the contact pads instead of a regular pushbutton. Adam and Sam also worked together on finalizing the activation, power drain, and recharge logic.



Our Arduino and breadboard setup.

Putting Everything Together

Once we had the circuit working, we started combining it with the physical ring. This ended up being one of the more difficult parts of the project. There wasn't much space inside the ring for the LED and wires, and the connections would sometimes come loose while we were trying to position everything. We ended up adding more tape inside the ring to help keep the wires in place. We also ran into problems with the foil contacts. Since we needed to prevent accidental activation, we decided to use a glove for insulation. However, the foil wouldn't stay attached very well with tape alone. We eventually used rubber bands to help secure it.

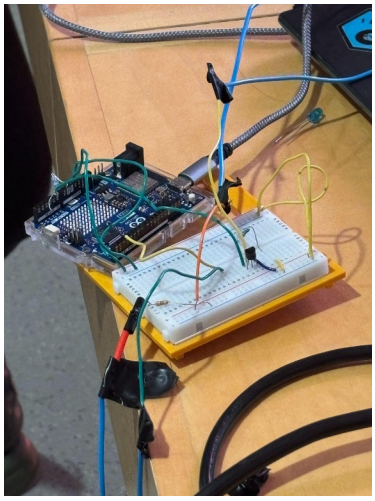


Testing the foil contacts and wiring.

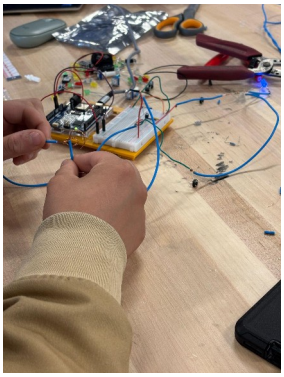
We spent quite a bit of time making small adjustments to the physical setup. During one session, we worked from around 7:30 to 11:50 trying to make the wiring more secure. Even when the circuit was working, moving the ring could cause a wire to come loose, so we had to be careful when testing it.

Testing and Results

Once everything was connected, we tested the contacts and checked how the LED responded. We were able to get the blue LED to light up during testing. We also worked on the fading and automatic recharge behavior so the ring could represent having a limited amount of power.



Our circuit during testing.



The blue LED lighting up during one of our tests.

Throughout the project, we took photos and recorded videos to document the different stages of construction. This included the early circuit, testing the contacts, and working with the physical ring.

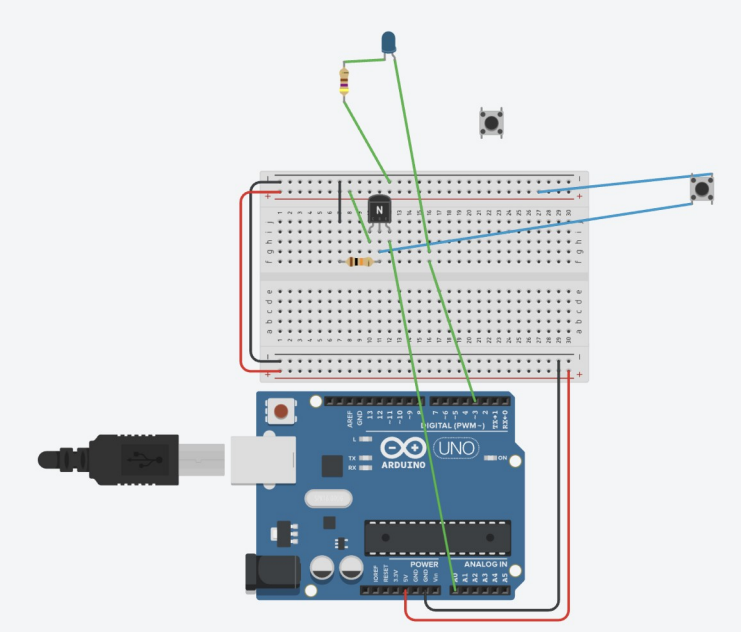
Final Demonstration

For our final demonstration, we recorded a video showing the ring and explaining the work that went into the project.

Our Project Runway ring demonstration.

Circuit Schematic

We used Tinkercad to plan and test the circuit before assembling the physical version. The circuit includes the Arduino, transistor, LED, resistors, and contact inputs.



Arduino Code

Our Arduino program handles the ring's activation, brightness, and recharge system. We used analogRead() to check the contact input at A0 and analogWrite() to control the LED's brightness. The program also keeps track of the ring's charge so the light can gradually dim and recharge after running out.

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Team Contributions

Adam - Ring Design and Physical Construction

Adam focused mainly on designing and building the physical ring. He researched different Blue Lantern ring designs, created the original sketch, and modeled the ring in Autodesk Fusion without using a premade model. He also completed the 3D-printing training, printed the ring, and made adjustments to the finished print so the wires could fit. A lot of his time went toward securing the wires, fitting the LED inside the ring, and getting the foil contacts to stay in place. He also worked with Sam on finalizing the logic for the activation, power drain, and recharge functions.

Sam - Circuit and Programming

Sam handled most of the circuit development and programming. He started with the transistor-based fading effect and used a Tinkercad example to help develop the initial circuit. He wrote the early code, worked on replacing the demo pushbutton with the contact-based input, and

used analog readings and outputs to control the LED. When the charging system wasn't resetting properly, he added a 10K pull-down resistor to stabilize the input. He also worked with Adam on the activation and recharge logic.

Brandon - Pseudocode and Documentation

Brandon worked on the pseudocode during the planning stage of the project. This involved laying out the basic logic for how the ring would activate, lose charge, and recharge over time. He also handled taking photos and recording videos throughout development. This included documenting the circuit, the physical construction, the testing process, and the final demonstration. Along with that, he was responsible for putting together the project documentation, organizing the photos and videos, and explaining the process and problems we ran into.

What We Learned

This project ended up taking way more work than we originally expected. At first, we thought getting the LED and Arduino working would be the biggest challenge, but a lot of our time went toward making the physical ring work properly. The 3D printing process was one example. Even though we planned space for the wires, we still had to make adjustments after printing because everything didn't fit the way we expected. We also learned how much small wiring problems can affect a project. The foil contacts needed to stay in position, the wires needed to remain connected, and the Arduino needed to read the input consistently. Adding the 10K resistor was a good example of how a small change could fix a problem we had been dealing with. If we had more time, we'd want to improve how the wires are secured and make the ring look more finished. We'd also like to find a cleaner way to attach the foil contacts to the glove. We enjoyed being able to combine programming, 3D modeling, and physical construction into one project. There was a lot of trial and error involved, but we learned quite a bit from having to figure out why things weren't working and making changes until they did.

References and Credits

- Arduino Language Reference - Arduino programming functions and LED control.
- Autodesk Tinkercad - Circuit planning and testing.
- Autodesk Fusion - Used to model the ring.
- DC Comics - Inspiration for the Blue Lantern ring design.
- RIT General Makerspace - 3D printing equipment and training.
- Physical Computing course materials - Circuit components and programming references.